MR3284880 (Review) 35B35 35B30 35B32 35K57 35L71

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Numerical results on existence and stability of steady state solutions for the reaction-diffusion and Klein-Gordon equations. (English summary)

Involve **7** (2014), *no. 6,* 723–742.

Summary: "In this paper, we study numerically the existence and stability of the steady state solutions of the reaction-diffusion equation, $u_t - au_{xx} - u + u^3 = 0$, and the Klein-Gordon equation, $u_{tt} + cu_t - au_{xx} - u + u^3 = 0$, with the boundary conditions: $u(-1) = u(1) = 0$. We show that as a varies, the number of steady state solutions and their stability change."